INYOUNG · CLASS 8

The Paper You'll Present

Four figures, four boxes. Then the whole talk, twice. Then the questions.

HOW TODAY WORKS

Four Boxes, Every Figure

St. Laurent et al., Biological Psychiatry 97:707 (2025) — ITC dysfunction and avoidance extinction in Sapap3 mice

1 ASKED

The question here is whether ___.

2 DID

To test this, they ___.

3 FOUND

What they found was ___.

4 SO WHAT

This tells us ___.

EXTINCTION 소거

Not forgetting. Learning something new: “the warning doesn't mean danger any more.”

잊는 게 아니라 , " 이제 그 신호가 위험을 뜻하지 않는다 " 를 새로 배우는 것 .

TENSE

What they did, what they found → past. They took the food away for six hours.

What it means → present. This tells us that hunger changes…

VOICE · out loud, active is easier to follow

✓ They switched the neurons off during the tone. △ The neurons were switched off during the tone. fine in a paper, harder to follow in a room

60 분 조립 · 35 분 통으로 두 번 · 25 분 질문 받기

PIECE 1 · THE BACKGROUND · 10 MIN

Why Ask This at All?

Why does a mouse keep climbing the platform when nothing bad happens any more?

SET IT UP IN THREE LINES

1. People with OCD keep doing something even when ___.
2. That looks like a failure of ___.
3. So the question here is whether ___.

YOUR LINE · 제일 잘 나온 문장 , 고친 버전

SAY IT RIGHT aMYGdala · obSESsive-comPULsive · exTINCtion · interCAlated

ITCd — 편도체 안의 작은 세포 묶음 . 주변을 끄는 역할을 해요 .

WHO

Sapap3-null mice — ___.

THE TASK

Sugar on one side, a safe platform on the other — ___.

WHAT GOES WRONG

In extinction, the tone comes without the shock, but ___.

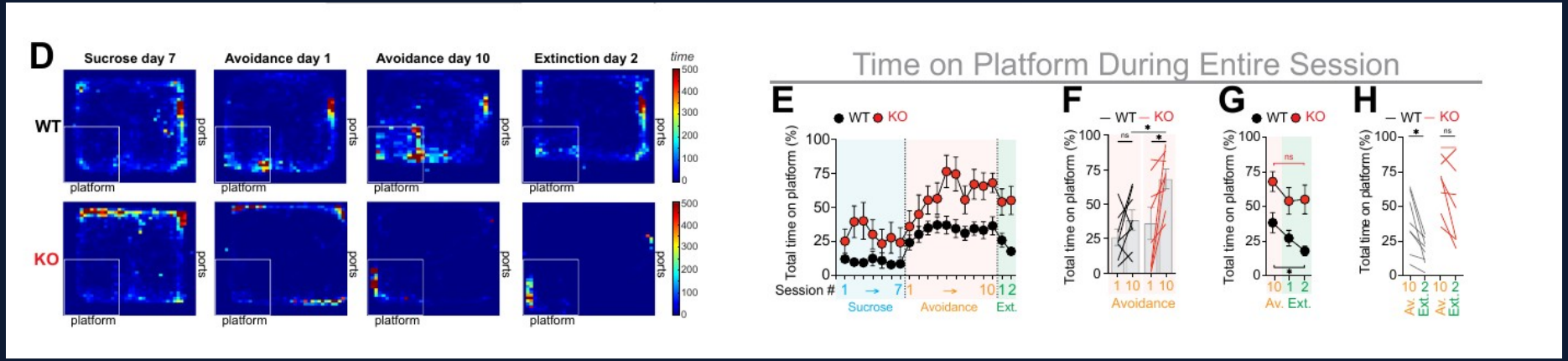
THE TARGET

ITCd — a small group of cells that ___.

PIECE 2 · FIGURE 1 · 12 MIN

What the Mice Did

How do you show that a mouse failed to extinguish — and not that it was simply more afraid?



St. Laurent et al., Biol Psychiatry 97:707 (2025) · Fig. 1D–H

1 ASKED The question here is whether ___.

2 DID To test this, they ___.

3 FOUND What they found was ___.

4 SO WHAT This tells us ___.

METHOD IN ONE LINE · heatmaps + time on platform

SAY IT TO UNDERGRADS 학부생한테 말하듯

platform-mediated avoidance → a choice between food and safety
extinction → the tone comes, the shock doesn't

HINT WORDS

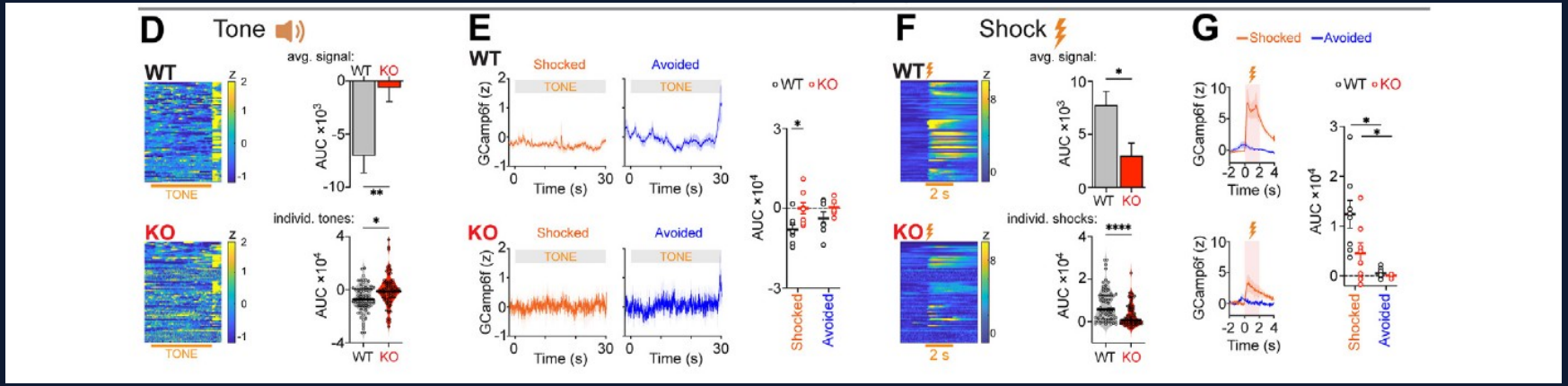
- platform-mediated avoidance
- 10 training days
- 2 extinction days
- KO stayed on the platform

SAY IT RIGHT aVOIDance · PLATform · deprivAtion · GENotype

PIECE 3 · FIGURE 2 · 15 MIN

What the Cells Did

If the behaviour is different, what does the ITCd signal look like while it happens?



St. Laurent et al., Biol Psychiatry 97:707 (2025) · Fig. 2D–G

1 ASKED The question here is whether ___.

2 DID To test this, they ___.

3 FOUND What they found was ___.

4 SO WHAT This tells us ___.

METHOD IN ONE LINE · how photometry works, in one sentence

SAY IT TO UNDERGRADS 학부생한테 말하듯

fiber photometry → watching a group of neurons light up while the animal moves

AUC → how much signal there was in total

HINT WORDS

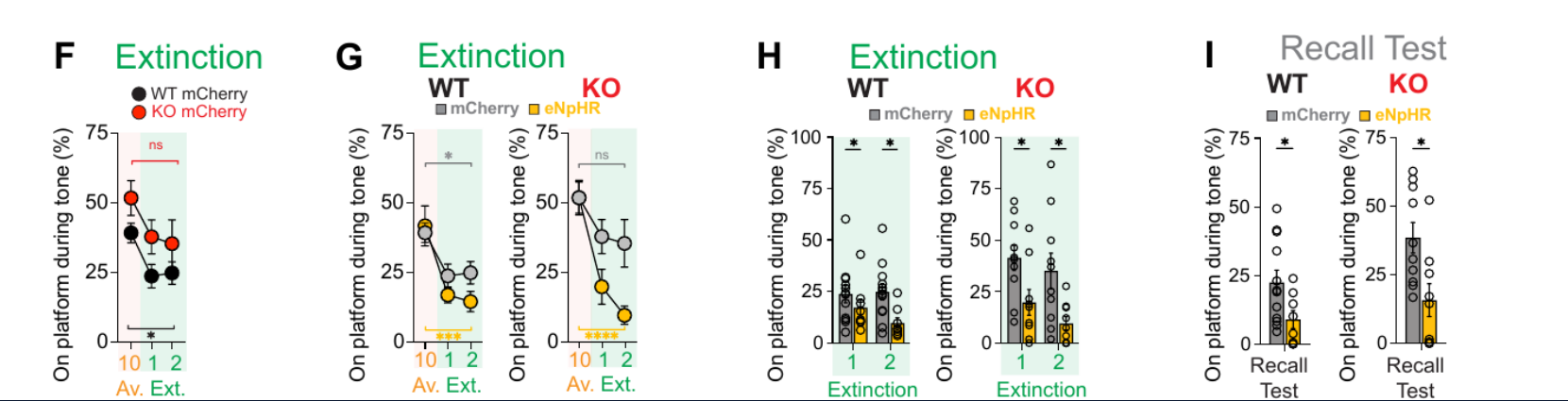
fiber photometry | GCaMP6f | tone vs. shock | AUC | weaker shock response

SAY IT RIGHT phoTOMetry · CALcium · TRANsient · AMplitude

PIECE 4 · FIGURE 3 · 15 MIN

Switch It Off

Is the ITCd signal just along for the ride, or is it causing the deficit?



St. Laurent et al., Biol Psychiatry 97:707 (2025) · Fig. 3F–I

1 ASKED The question here is whether ___.

2 DID To test this, they ___.

3 FOUND What they found was ___.

4 SO WHAT This tells us ___.

METHOD IN ONE LINE · when exactly the light was on

SAY IT TO UNDERGRADS 학부생한테 말하듯

optogenetic inhibition → switching the cells off with light
recall test → testing again a day later, with no light

HINT WORDS

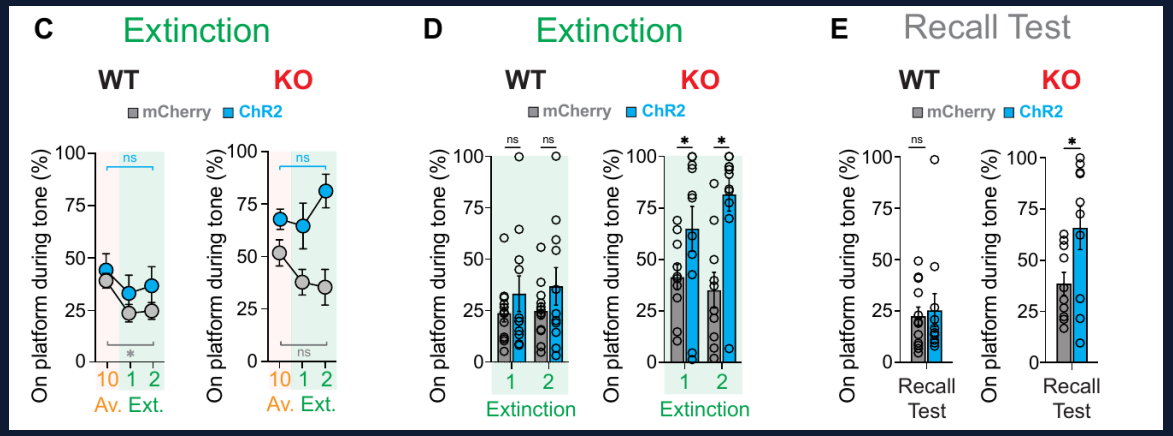
- eNpHR3.0
- light during the 30-second tone
- extinction rescued
- recall test 24 h later

SAY IT RIGHT optoGEnetic · inhiBItion · haloRHOdopsin · reCALL

PIECE 5 · FIGURE 4 · 12 MIN

Switch It On

And the other direction — what happens if you drive the same cells harder?



St. Laurent et al., Biol Psychiatry 97:707 (2025) · Fig. 4C–E

1 ASKED The question here is whether ___.

2 DID To test this, they ___.

3 FOUND What they found was ___.

4 SO WHAT This tells us ___.

METHOD IN ONE LINE · what the control group got

SAY IT TO UNDERGRADS 학부생한테 말하듯

ChR2 activation → switching the cells on with light
mCherry control → the light, but no switch

HINT WORDS

ChR2 | KO got worse | WT: no effect | persisted to the recall test

SAY IT RIGHT actiVAtion · channelrhoDOPsin · exACerbate · perSISted

PIECE 6 · THE WHOLE STORY · 35 MIN

Start from the Top

두 번 해요 . 두 번째는 멈추지 않고 , 그림을 가리키면서 .

FIG 1 What the mice did — ___.

FIG 2 What the cells did — ___.

FIG 3 Switch it off — ___.

FIG 4 Switch it on — ___.

TAKE 2 · 두 번째에 더 좋아진 한 줄

1 분 배경 → Fig 1 → 2 → 3 → 4 → 마무리 한 줄 .

CLOSE IT · 하나 고르기

Mechanism

So the ITCd has to quiet down for the animal to let go of the habit — and in these mice, it doesn't.

Big picture

This is one of the first times a single ITC cluster has been switched off, and the behaviour followed.

For the room

It points at a circuit you could actually aim a treatment at.

PIECE 7 · WHEN THEY ASK · 25 MIN

The Questions You'll Get

Brian 이 journal club 처럼 물어요 . 첫마디부터 .

1. How did you know the mouse was avoiding, and not just sitting there?
2. Why only the dorsal cluster? What about the ventral one?
3. Two days of extinction — is that long enough?
4. Why did the activation do nothing in the wild-type mice?
5. How long does one photometry session take?
6. If you ran the next experiment, what would it be?

FIRST MOVES 첫마디

- That's a good question — let me go back to Figure ~.
- I'm not sure, but my guess is ~.
- The short answer is ~, and the longer answer is ~.
- We didn't test that, but ~.
- I'd have to check the methods, but ~.

THE ONE YOU'LL REDO · 다시 답하고 싶은 것

FROM YOUR WORDS · 이번 주

speculate | improvise | reiterate

WRAP

Before the Lab Meeting

오늘 나온 문장이 L&S 03 으로 가요 .

1

Record it

The three-minute version — about 40 seconds per figure. Send it to Brian.

3 분 버전을 녹음해서 보내기

2

Find out

How long the talk is · the date · whether the slides are yours or your professor's.

발표 길이 · 날짜 · 슬라이드 확인

3

Listen

L&S 03 is built from today's YOUR LINE sentences — your own words, corrected.

오늘 나온 문장으로 만든 복습 자료